



NATIONAL LEVEL HACKATHON 2026 • HEALTHCARE & AI TRACK

ArogyaGram

Intelligent Community Health-Response Network

“Turning unstructured human distress into coordinated real-world medical action in under 40 seconds.”

 Multilingual AI Triage

 Deterministic 108 SOS

 Verified Responders

 Smart Prescription Care

The 40–Kilometer Healthcare Desert

65%

Rural Population

Over 650M Indian citizens reside in rural panchayats with access to under 30% of emergency healthcare resources.

90+ Min

Ambulance Transit Lag

Average emergency transit delay across Tier-3 panchayats, hilly terrain, and remote homesteads.

1 : 10,000

Doctor Deficit

Severe rural practitioner deficit in community clinics compared to the WHO recommended 1:1,000 ratio.

The Daily Ground Reality in Isolated Panchayats:

- **Chronic Neglect:** Elderly diabetic patients miss insulin & BP doses because pharmacies are 10–15 km away.
- **Unwitnessed Emergencies:** Falls, cardiac events, and domestic trauma go unreported for critical golden hours.
- **ASHA Isolation:** Health workers manage acute clinical events without real-time diagnostic decision support.
- **Linguistic Barriers:** Inability to convey dialect symptom histories in standardized terminology.

ArogyaGram: One Unified Platform, Four Pillars



Clinical AI Triage (Ask Arogya)

- Powered by Google Gemini 2.5 Flash + Deterministic Clinical Rules
- Native voice-to-text in Malayalam, Hindi & English via Capacitor
- Evaluates urgency: LOW, MODERATE, HIGH, CRITICAL risks
- Floating zero-overlap input card + automatic reading scroll



Deterministic Emergency SOS

- Automatic red-flag trauma detection bypasses AI for immediate rescue
- 1-Tap direct emergency dial to 9539141210 / 108 Ambulance
- Instant GPS coordinate capture with reverse geo-tagging
- Multi-recipient caregiver alert dispatch with live status



Volunteer Task & Responder Network

- Mobilizes verified local neighbors for high-impact care tasks
- 100-Point algorithmic matching (distance, skills, rating, vehicle)
- Interactive Leaflet map view for active community assistance calls
- Scoped non-clinical tasks: medicine pickup, hospital escorts



Structured Medication Adherence

- Replaces forgotten paper prescriptions with digital schedules
- Structured dosage picker: tablets, syrups, strength & meal relations
- Native 24h clock selector + fast mealtime presets (M/N/E)
- Strict 10-digit mobile number rules & Supabase cloud sync

From Colloquial Distress Signal to Coordinated Action

STEP 01

Native Voice Intake

Citizen speaks in dialect (Malayalam/Hindi) or types raw symptoms. Offline-ready Android speech engine captures transcript.

Zero medical jargon required

STEP 02

Situation & Safety

Gemini triages distress severity. Trauma red flags trigger instant SOS bypass; stable needs create structured Care Cases.

Deterministic triage guardrails

STEP 03

Volunteer Dispatch

MatchingService scores nearby responders (0–100) based on distance, verified skills, and transport. Best neighbor alerted.

Algorithmic neighbor routing

STEP 04

Resolution & Trail

Task verified on ground. Complete audit history logged to Unified Health Timeline. Encrypted Supabase backup.

Audited, closed-loop care

Gemini Intelligence with Deterministic Safety Bounds

Clinical AI & Safety Guardrails

- Multilingual NLP:** Gemini 2.5 Flash natively processes Malayalam, Hindi, and English with clinical grounding.
- Triage Classification:** Categorizes urgency into LOW, MODERATE, HIGH, CRITICAL with recommended escalation.
- Configurable API Key:** Dedicated in-app settings screen to test, validate, and store custom Gemini keys locally.
- Deterministic Safety Bypass:** Acute symptoms bypass AI to eliminate critical response latency.
- Strict Volunteer Scoping:** Volunteers perform ONLY logistics and accompaniment, never medical acts.

Production Engineering Stack

- Core Framework:** React 19 + TypeScript + Vite (2,077 modules, ~1.3s build)
- Native Packaging:** Capacitor 6 + Speech Recognition + Custom App Branding
- Intelligence:** Google Gemini 2.5 Flash SDK + Situation Engine
- Geospatial:** Leaflet.js + OpenStreetMap for live volunteer dispatch
- Database:** Supabase (PostgreSQL with RLS & real-time sync)
- Validation:** Strict 10-digit Indian phone verification + live counters

Real-World Grounding: Kanjirappally Block, Kottayam

Mapped Healthcare Ecosystem

MaryQueens Mission Hospital

Multispecialty tertiary referral hospital with 24/7 emergency trauma care and ICU backup.

Kanjirappally General Hospital CHC

Public community health centre providing free diagnostic testing, fever clinics & pharmacy refills.

Chirakkadavu PHC

Sub-block immunization clinic, ASHA hub, and maternal health monitoring center.

St. Dominic's 24/7 Pharmacy

Emergency nocturnal medicine dispensary stocked with cardiovascular & diabetic medications.

< 40 Secs

Rapid Care Dispatch

From voice distress to nearby verified responder alert.

100%

Offline Resilience

Zero downtime: cached triage & local emergency contacts.

Empowering Frontline ASHA Workers

Instead of tedious paper logs, health workers get automated triage and emergency escalation in their pocket.

180 ASHA Hubs

Tested and Verified on Physical Android Hardware



Native Android APK

- Packaged with Capacitor 6
- Verified on Redmi Note 14 5G
- Official adaptive launcher icons
- Lightweight ~4.1 MB download size



Native Voice-to-Text

- Malayalam, Hindi, and English
- On-device speech recognition
- One-tap microphone controls
- Instant transcription into chat



Supabase & Data Sync

- Live community health news
- Verified Kanjirappally clinics
- Resilient offline fallback cache
- Strict 10-digit phone verification

On-Device Performance Benchmarks (Redmi Note 14 5G):

Cold Launch: < 850 ms to full interactive home screen

Voice Latency: Sub-second speech-to-text response

Chat Layout: Pinned floating card above navigation

Build Speed: ~400 ms incremental Gradle compilation



NATIONAL LEVEL HACKATHON 2026 • FINAL PITCH

“No Rural Citizen Should Face Illness Alone.”

ArogyaGram doesn't wait for physical infrastructure to catch up. It bridges the 40-km healthcare desert using the smartphone in every villager's pocket.

The Ask

Partnership with National Health Mission (NHM) Kerala to pilot across 4 rural panchayats in Kottayam district.

Next Steps

ASHA worker beta group onboarding, Supabase database sync, and ABDM health ID integration.

Live & Functional

Verified Android APK build available.
Complete source code on GitHub:
github.com/jvstin47/ArogyaGram-3